any intercellular space. Epidermal pores and cuticle are absent in the piliferous layer. Root hairs that are found in the piliferous layers are always unicellular. They absorb water and mineral salts from the soil. Root hairs are generally short lived. The main function of piliferous layer is protection of the inner tissues.

### Cortex

The cortex is homogenous. i.e. the cortex is made up of only one type of tissue called parenchyma. It consists of many layers of thin-walled parenchyma cells with lot of intercellular spaces. The function of cortical cells is storage. Cortical cells are generally oval or rounded in shape. Chloroplasts are absent in the cortical cells, but they store starch. The cells are living and possess **leucoplasts.** The inner layer of the cortex is endodermis. It is composed of single layer of barrel shaped parenchymatous cells. This forms a complete ring around the stele. There is a band like structure made of **suberin** and **lignin** present in the radial and inner tangential walls of the endodermal cells. They are called **casparian strips** named after **casparay** who first noted the strips. The endodermal cells, which are opposite the protoxylem elements, are thin walled without casparian strips. These cells are called passage cells. Their function is to transport water and dissolved salts from the cortex to the xylem. Water cannot pass through other endodermal cells due to casparian strips. The main function of casparian strips in the endodermal cells is to prevent the re-entry of water into the cortex once water entered the xylem tissue.

### Stele

All the tissues inside the endodermis comprise the stele. This includes pericycle, vascular system and pith.

### Pericycle

Pericycle is the outermost layer of the stele and lies inner to the endodermis. It consists of single layer of parenchymatous cells.

### Vascular System

Vascular tissues are seen in radial arrangement. The number of protoxylem groups is many. This arrangement of xylem is called polyarch. Xylem is in exarch condition, the tissue which is present between the xylem and the phloem, is called conjunctive tissue. In maize, the conjunctive tissue is made up of sclerenchymatous tissue.

**Anatomical differences between dicot root and monocot root**

| S.No. | Characters | Dicot root | Monocot root |
|---|---|---|---|
| 1. | Pericyle | Gives rise to lateral roots, phellogen and a part of vascular cambium. | Gives rise to lateral roots only. |
| 2. | Vascular tissue | Usually limited number of xylem and phloem strips. | Usually more number of xylem and phloem strips, |
| 3. | Conjunctive tissue | Parenchymatous; Its cells are differentiated into vascular cambium. | Mostly sclerenchymatous but sometimes parenchymatous. It is never differentiated in to vascular cambium. |
| 4. | Cambium | It appears as a secondary meristem at the time of secondary growth. | It is altogether absent. |
| 5. | xylem | Usually tetrach | Usually polyarch |



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